

Xinyu Shi

Ph.D. Candidate in School of Computer Science, University of Waterloo

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RESEARCH INTERESTS

My research focuses on **HCI** and **Human-Centered AI** for open-ended creative tasks, aiming to *build AI systems that align with individual users' goals and values*. My work spans two connected threads: (1) **designing new human-AI interaction paradigms** — externalizing ambiguous human intent for creative artifacts creation and making model behaviors more inspectable and steerable; and (2) **integrating human insight into AI** — translating the tacit domain knowledge and human visual tastes into computational representations for model training and fine-tuning.

EDUCATION

- 2021 – Present **Ph.D. in Computer Science**
University of Waterloo, Canada
Advisor: Prof. Jian Zhao
- 2016 – 2020 **B.Eng. in Software Engineering**
Xiamen University, China
Advisor: Prof. Shihui Guo

EMPLOYMENT

- Summer 2025 **Adobe Research** — Creative Technologies Lab
Research Scientist Intern
Hosts: Dr. Rubaiat Habib Kazi, Dr. Li-Yi Wei, Dr. Cherry Zhao
Redefined how creators interact with, understand, and supervise VLMs in animation creation [P9].
- Summer 2024 **Adobe Research** — AI Experience Lab
Research Scientist Intern
Hosts: Dr. Shunan Guo, Dr. Jane Hoffswell, Dr. Gromit Chan, Dr. Eunyee Koh
Modeled abstract design intent into computational representations for infographic generation [A2].
- Summer 2023 **Microsoft Research** — Data, Knowledge & Intelligence Group
Research Scientist Intern
Host: Dr. Yun Wang
Repurposed visualization as the instrument to express and refine color intent in video [P6].

PEER-REVIEWED PUBLICATIONS (FULL PAPERS)

Google Scholar: [link]; Semantic Scholar: [link]

Note about venues: CHI is considered as the top-tier, most prestigious venues in the field of Human-Computer Interaction (HCI), with an annual acceptance rate of 20 - 25%.

🏆 Best Paper Award; 🏅 Honorable Mention and Other Awards; * Equal Contribution.

- [P9] CHI 2026 **Xinyu Shi**, Li-Yi Wei, Nanxuan Zhao, Jian Zhao, Rubaiat Habib Kazi. Notational Animating: An Interactive Approach to Creating and Editing Animation Keyframes. *In Proceedings of the ACM Human Factors in Computing Systems (CHI)*. 2026.
🏅 Best Paper Honorable Mention.
- [P8] ACL 2026 Zijian Chen*, Xueguang Ma*, Shengyao Zhuang*, Ping Nie, Kai Zou, Andrew Liu, Joshua Green, Kshama Patel, Ruoxi Meng, Mingyi Su, Sahel Sharifymoghaddam, Yanxi Li, Haoran Hong, **Xinyu Shi**, Xuye Liu, Nandan Thakur, Crystina Zhang, Luyu Gao, Wenhui Chen, Jimmy Lin. BrowseComp-Plus: A More Fair and Transparent Evaluation Benchmark of Deep-Research Agent. *In Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*. 2026.
- [P7] CHI 2025 **Xinyu Shi**, Yinghou Wang, Ryan Rossi, Jian Zhao. Brickify: Enabling Expressive Design Intent Specification through Direct Manipulation on Design Tokens. *In Proceedings of the ACM Human Factors in Computing Systems (CHI)*. 2025.
- [P6] CHI 2024 **Xinyu Shi**, Yinghou Wang, Yun Wang, Jian Zhao. Piet: Facilitating Color Authoring for Motion Graphics Video. *Proc. ACM Human Factors in Computing Systems (CHI)*. 2024.
🏆 Best Paper Award.
- [P5] CHI 2024 **Xinyu Shi**, Mingyu Liu, Ziqi Zhou, Ali Neshati, Ryan Rossi, Jian Zhao. Exploring Interactive Color Palettes for Abstraction-Driven Exploratory Image Colorization. *In Proceedings of the ACM Human Factors in Computing Systems (CHI)*. 2024.
- [P4] CHI 2023 **Xinyu Shi**, Ziqi Zhou, Julia Zhang, Ali Neshati, Anjul Kumar Tyagi, Ryan Rossi, Shunan Guo, Fan Du, Jian Zhao. De-Stijl: Facilitating Graphics Design with Interactive 2D Color Palette Recommendation. *In Proceedings of the ACM Human Factors in Computing Systems (CHI)*. 2023.
- [P3] MM 2022 **Xinyu Shi***, Chaofeng Chen*, Yipeng Qin, Xiaoming Li, Xiaoguang Han, Tao Yang, Shihui Guo. Real-World Blind Super-Resolution via Feature Matching with Implicit High-Resolution Priors. *In Proceedings of the ACM Multimedia (MM)*. 2022.
🏅 Oral Presentation (Top 5%).
- [P2] Ubicomp 2021 Qijia Shao, Amy Sniffen, Julien Blanchet, Megan E. Hillis, **Xinyu Shi**, Themistoklis Haris, Jason Liu, Melissa Malzkahn, Lorna Quandt, James Mahoney, David J.M. Kraemer, Xia Zhou, Devin Balkcom. Teaching American Sign Language in Mixed Reality. *In Proceedings of the ACM Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT, presented in Ubicomp)*. 2021.

- [P1] **Xinyu Shi**, Junjun Pan, Zeyong Hu, Juncong Lin, Shihui Guo, Minghong Liao, Ye Pan, ISMAR 2019 Ligang Liu. Accurate and Fast Classification of Foot Gestures for Virtual Locomotion. *In Proceedings of the IEEE Intl. Symposium on Mixed and Augmented Reality (ISMAR). 2019.*

IN SUBMISSIONS

- [S3] Crystina Zhang, Zigeng Xu, **Xinyu Shi**, Raphael Tang, Nandan Thakur, Xueguang Ma, Under Review Yao Lu, Jimmy Lin. Hawkeye: Inspecting Agentic Search Trajectories. *In Submission.*
- [S2] **Xinyu Shi**, Chun Ye, Shunan Guo, Jane Hoffswell, Gromit Chan, Eunyee Koh, Jian Zhao. Under Review Diagrammatic Sketching: Exploring Infographic Design Alternatives. *In Submission.*
- [S1] Cara Li, Xiang Li, **Xinyu Shi**, Jian Zhao. Semantic Ink: Supporting Reuse with Variation Under Review in Annotational Graphics Authoring. *In Submission.*

EXTENDED ABSTRACTS, WORKSHOPS

- [A2] **Xinyu Shi**, Shunan Guo, Jane Hoffswell, Gromit Yeuk-Yin Chan, Victor S. Bursztyn, Jian CHI EA 2025 Zhao, Eunyee Koh. Comprehensive Sketching: Exploring Infographic Design Alternatives in Parallel. *Extended Abstracts of ACM Human Factors in Computing Systems (CHI Late-Breaking Work 2025).*
- [A1] Zijian Chen*, Xueguang Ma*, Shengyao Zhuang*, Ping Nie, Kai Zou, Andrew Liu, Joshua NeurIPS 2025 Green, Kshama Patel, Ruoxi Meng, Mingyi Su, Sahel Sharifymoghaddam, Yanxi Li, Haoran MTI-LLM Hong, **Xinyu Shi**, Xuye Liu, Nandan Thakur, Crystina Zhang, Luyu Gao, Wenhui Chen, Jimmy Lin. BrowseComp-Plus: A More Fair and Transparent Evaluation Benchmark of Deep-Research Agent *NeurIPS 2025 MTI-LLM Workshop*. 🏆 **Spotlight**.

PAPER AWARDS

- 2026 **Best Paper Honorable Mention Award** at CHI 2026 as first author [P9]
Awarded to the top 5% of submissions made to the CHI conference.
- 2025 **Spotlight** at NeurIPS MTI-LLM Workshop as co-author [A1]
Awarded to the top 7% of accepted submissions to NeurIPS MTI-LLM Workshop.
- 2024 **Best Paper Award** at CHI 2024 as first author [P6]
Awarded to the top 1% of submissions made to the CHI conference.
- 2022 **Oral Recognition** at ACM MM as co-first author for FeMaSR [P3]
Recognition given to the top 5% of submissions made to the MM conference for oral presentation.

HONORS AND SCHOLARSHIPS

- 2026 **Jane Street Fellowship, Honorable Mention**
One of the 28 Honorable Mention Recognitions among the world-wide competitive applicant pool.

- 2025 – 2027 **David R. Cheriton Graduate Scholarship**, University of Waterloo
Awarded to top PhD students based on academic research excellence. \$20,000 over 2 years.
- 2022 – 2026 **International Doctoral Student Award**, University of Waterloo
Awarded to top international doctoral students based on academic excellence. \$53,000 over 4 years.
- 2024 – present **Special Recognitions for Outstanding Reviews** (5 in total)
Recognition to outstanding reviewers provided highly useful peer reviews.
- 2021 **Vector Scholarship in AI**, Vector Institute
Awarded to top incoming graduate students for research potential in AI. \$17,500 over 1 year.
- 2021 **Mathematics International Award of Excellence**, University of Waterloo.
Awarded to incoming international graduate students for research potential. \$5,000 over 2 terms.
- 2016 – 2020 **Award for Excellence in Academic Performance**, Xiamen University
Awarded to top undergraduate students based on academic excellence. 5,000 CNY each time.

PAST RESEARCH POSITIONS

- 2021/09 – **University of Waterloo**, Waterloo, ON, Canada
Present Graduate Student Researcher
Advisor: Jian Zhao
Building interactive systems to augment human creativity [P4, P5, P7].
- 2020/12 – **The Chinese University of Hong Kong, Shenzhen**, Shenzhen, China
2021/08 Research Assistant
Advisor: Xiaoguang Han
Proposed an approach based on VQ-GAN for Image Super-Resolution [P3].
- 2019/09 – **Dartmouth College**, Hanover, NH, USA
2020/03 Research Assistant
Advisors: Xia Zhou, Xing-Dong Yang
Leveraged wearable devices & Mixed Reality for Sign-Language teaching [P2].
- 2017/10 – **Xiamen University**, Xiamen, China
2019/09 Undergraduate Research Assistant
Advisor: Shihui Guo
Exploited wearable devices for VR locomotion with gesture recognition [P1].

ACADEMIC SERVICES

Organizing Committee

- 2025 – 2026 Assistant to Paper Chairs, CHI 2026

Program Committee

- 2025 International Program Committee, GI 2025
2025 Program Committee, Associate Chair, C&C 2025

Conference Session Chair

- 2026 Paper Session “*Role Play, Creativity and AI*”, CHI 2026
- 2025 Paper Session “*Malleable and Adaptive Interface*”, CHI 2025

Reviewer

- 2024 – 2026 ACM User Interface Software and Technology Symposium (UIST)
- 2024 – 2025 SIGCHI Conference on Human Factors in Computing Systems (CHI)
- 2026 IEEE Transactions on Visualization and Computer Graphics (TVCG)
- 2024 – 2025 IEEE Visualization and Visual Analytics Conference (VIS)
- 2024 – 2025 SIGCHI Late Breaking Work (CHI LBW)
- 2025 Pacific Conference on Computer Graphics and Applications (Pacific Graphics)
- 2024 ACM Conference on Designing Interactive Systems (DIS)
- 2024 Creativity & Cognition (C&C)

🏆 Special Recognitions for Outstanding Reviews (×5):

CHI’24, CHI’25, UIST’25, CHI LBW’25, C&C’25

Student Volunteer

- 2026 Graphics Interface (GI)

INVITED TALKS

Interactive Visual Abstractions for Computational Creativity

- 2026/05 Guest Lecture for CS449 at the University of Waterloo, hosted by Prof. Jian Zhao.
- 2026/01 Seminar at the CUHK (Shenzhen), School of AI, hosted by Prof. Feng Yin.
- 2025/11 University of Toronto iSchool Seminar, hosted by Prof. Ebrahim Bagheri.
- 2025/11 Guest Lecture for CS889 at the University of Waterloo, hosted by Prof. Jian Zhao.
- 2025/11 UCSD Design Lab Seminar, hosted by Sirui Tao and Zhicheng Huang.

MENTORING

- 2025 – **Cara Li**, Master student at University of Waterloo.
Project: Fluid in-situ graphics creation through reified sketched annotations [S1].
- 2024 – 2025 **Chun Daisy Ye**, Undergraduate student at University of Waterloo, published [S2].
Now: Software Engineer at Salesforce.
- 2023 – 2024 **Yinghou Wang**, Master student at Harvard University & MIT, published [P6, P7].
Now: Creative Technologist at Motorola Solutions.
- 2023 **Mingyu Liu**, Undergraduate student at University of Waterloo, published [P5].
Now: Software Engineer at Bloomberg.
- 2022 **Ziqi Zhou**, Master student at University of Waterloo, published [P4].
Now: Partner at Puzzle Ventures.
- 2022 **Julia Zhang**, Undergraduate student at University of Waterloo, published [P4].
Now: Software Engineer at Whatnot.

TEACHING EXPERIENCE

Teaching Assistant in University of Waterloo

- CS246 Object-Oriented Software Development. (Winter 2022, Fall 2023, Winter 2025)
- CS251 Computer Organization and Design. (Spring 2022, Winter 2023)
- CS234 Data Types and Structures. (Fall 2022)
- CS115 Introduction to Computer Science. (Fall 2021)

PRESS COVERAGE

- 2024 *Waterloo News*: Scheming colours faster and easier. [[Link](#)]
- 2023 *GraphicSpeak*: University of Waterloo's Computer Scientists want Graphic Designers to use better Colors. [[Link](#)]
- 2023 *TechTimes*: Computer Scientists Develop "De-Stijl" Tool With Adobe to Help People Use Color Better in Graphic Design. [[Link](#)]
- 2023 *Waterloo News*: Dreaming in technicolour: Waterloo computer scientists develop a tool to help people use colour better in graphic design. [[Link](#)]